

Emerging Technology in Public Safety

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An emerging technology that I have identified that can be adopted by the Sheriff's office are artificial olfactory systems into robots, allowing them to have a sense of smell by mimicking wildlife. Living organisms have olfactory receptors and proteins that molecules from their surroundings bind to; this binding sends a neural signal to the brain, and the smell is identified (Wang et al). The technology currently being researched have the potential to not only determine what the smell is, but also where it is being emitted from, requiring more advanced sensing frameworks and carefully synthesized materials achieve these goals.

It was thought that investigating molecular properties through chemosensors, sensors that change a signal from tiny molecular interactions, and gas chromatography, the method of separating mixtures into molecules using gases and liquids, was the closest technology available for the job (Lazar). However, while separating molecules is an important capability, this analysis is limited to their chemical properties and the number of particles within the sample to identify their compound structure. Today, there are many gas sensors that are developed to be sensitive to different molecules, each with a different set of advantages and weaknesses. But, if the correct ones are chosen based on determined benchmarks and are integrated together into a singular calculating system, then the effectiveness to analyze smells is enhanced.

A metal oxide semiconductor (MOS) gas sensor measures changes in resistance when a target gas is detected, making it a chemiresistive sensor. They have a high sensitivity compared to other gas sensors, allowing for quick detection and smell identification, and this can be increased further by doping with different metals to shift its sensitivity towards specific groups of molecules (Wang et al). A drawback of this high sensitivity is their poor selectivity, making it

harder to identify the molecule exactly. However, this can be improved by installing filtration materials that allow molecules of certain sizes to pass through or adding a catalyst that starts a reaction to decompose the target molecule into many smaller ones (Wang et al).

Another gas sensor suitable for artificial smelling is a field-effect transistor (FET) gas sensor, which are similar to MOS sensors by their compact size, high sensitivity, and low cost. A FET works using two conductive materials, a source and a drain, controlled by a gate terminal connected to a sensing material (NIH). This gate sits on top of the semiconductive material between the source and drain to allow or block electron flow. When a target molecule is detected, it sends a voltage signal to the gate, changing the current across the source and drain and thus changes conductivity, or the amount of charge being flowed across. FET sensors continue to become optimized thanks to research into 1D and 2D nanostructured materials to diversify the range of sensed molecules by a larger surface area-volume ratio (Middya et al.) They are more advantageous than MOS sensors because they sense using more parameters and thus allowing more control in its measurement, but they can be used together to produce precise information about the smell.

These sensors can be used to detect sources of danger or the concentration of the surrounding environment to promote the safety of civilians within the concerned area. For instance, they could sense food and air contamination, detect diseases from emitted body gases, or identify gas leakages and predict gas explosions (Lazar). However, an unmet need of the sheriff's office is the immediate identification of select drugs or explosives such as bombs and firearms without risking the life of a human or a trained animal. If there is a building or property that is known to possess these hazardous items, then an army of smelling robots that are each

sensible to different molecules can give the office useful information about the in-house substances in seconds, potentially saving many lives.

The office can request for many small drones or robotic dogs each sensible to specific molecules and have the capability to apply different filtration materials or software that help recalibrate the sensors to detect different particles. A specific example the sheriff's office can use them for is a responding to a call about suspected explosives such as following a bomb threat. A key compound that is used in dynamite is nitroglycerin, which is known to have a low vapor pressure, meaning that it emits few molecules and thus making it hard to smell (gHale). If a human or K-9 enters the facility and does not detect the explosives, then there is a chance they trigger a catastrophe before they would be found. However, several sensitive robots can enter the premise and immediately detect the compound from several feet away and therefore it can be diffused or be detonated safely in a more secluded area. Moreover, they can be equipped with cameras that send live video to the sheriff's office; if a suspect directly interferes with a robot's operations, then it would catch the person's appearance and reduce the work of investigators and forensic teams.

This technology can be implemented without constructing any new buildings; however, it may require several specialized rooms in existing ones to keep the drones and robotic dogs in storage or training. There may be designated launching docks like the existing ones for active drones located on rooftops or in private parking areas, which are assigned as checkpoints spread across the county. At each checkpoint, there may be a laboratory room for repairs, calibration, or inspections for all units to ensure they are ready for their assigned task. With these considerations for new infrastructure and resources, the total cost to service the robots is not substantial, however it must fulfill the need to strengthen the public safety within surrounding

communities. But, adding a new unit to the sheriff's office will demand many new technologies on top of new robots such as new computers, monitors, power supplies, and virtual reality tools. The installation of new infrastructure is done so that the sheriff's office is only concerned about whether the robots work or not without adding knowledge debt about chemistry and electronics.

All introductory training, which emphasizes learning about different mixtures, their chemical properties, classifications, and reading safety labels can be done inside the police academy or within the sheriff's office with a low demand for infrastructure. Then, all trainees must familiarize with the sensors being used on duty and associate the molecules to be sensed with the chemicals that filter them for each sensor. This is done with minimal chemistry knowledge necessary for the trainees since this burden is handled by the engineers preparing and testing the robots. Finally, any training on flying the drones or maneuvering the robotic dogs are done through virtual simulations either inside the police academy or at the assigned checkpoints where they are deployed from. The trainee is not required to memorize the periodic table, recite chemical reactions, or investigate the calculations done to compute significant chemical properties since the software handles the processing of target molecules to output its underlying characteristics.

There are additional risks and concerns that come with installing the new "smelling" technology because of the many safety precautions that must be met when handling dangerous chemical products or transporting the robots into an unsafe area. To begin, any facility that stores chemical products must follow OSHA guidelines and install new emergency equipment and stations. All robotic systems may be surveilled and maintained in the same manner as body cameras and existing drone units, adding another set of liabilities to exhaust money and resources on for training, part and unit replacement. As with all software products, the robots

may be subjected to malware attacks if their cybersecurity measures are not maintained. Any situation where all power supplies are out of service, or the software is offline, may require intervention from agencies or departments with the technology to inspect these substances. The robots would not replace existing units within the sheriff's office, although their intent to be adapted at the county level is to increase the efficiency to identify the chemicals in order to implement ways to remove them off life-threatening areas. In other words, the technology would not be designed to decontaminate or safeguard locations where these hazards are identified because they only return information about the chemical makeup of the interested environment.

If I were advising the sheriff's office, then I would recommend them to examine the idea of artificial smell because of the rapid advancement in existing sensing devices. It brings the sensors popular in research and industrial applications into a radical concept, and specialized programs must be developed that translates electrical sensor output into a quantifiable representation of a separated mixture sample. However, without a software application that combines chemical makeup with robotic control, the office cannot implement the technology yet. The application is its own project that must build on top of decades of research into fabrication techniques and materials, and it must integrate a program and hardware structure that has been realized within the past decade (Lazar). To reiterate, more progress must be done in system design before the idea can realistically be applied outside of research. But because the sensing and robotic technologies already exist, conceiving the idea becomes less challenging if more public safety bodies become interested and have researchers advise them on implementing it. I feel optimistic that robots imitating the sense of smell promotes the health and safety of everyone involved because of their task of exploring hazardous areas while civilians and sheriff's office staff remain clear of danger.

Works Cited

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